



STUDENT

Ankit Sharma

ASSIGNMENT

NOF1 - QA NOF1

TOPIC

**Functions: Mappings, One-One,
Onto, Bijection**

EVALUATION DATE

14 May 2026**76.0%**

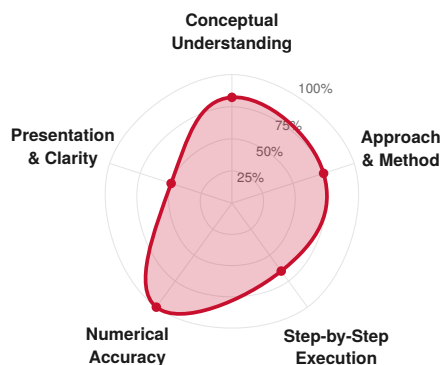
4.56 / 6 marks

Trailblazer

Qualifies for AA

PROMOTION: Promoted to AA.Threshold ($\geq 75\%$) cleared by 1.0 percentage points.**SUMMARY**

Ankit got all six final answers correct on this warm-up - a strong outcome on paper. The concern is underneath: on Q1(B) and Q5, the reasoning that produced the correct letter is itself flawed, and on Q3 a 0/0 slip nearly derailed the argument. The next focus area is tightening proofs and using the simplest available tool rather than over-engineered case analysis.

Rubric Breakdown

Dimension	Wt	Score	Status
Conceptual Understanding	40%	82.3%	Strong
Approach & Method	20%	75.0%	Strong
Step-by-Step Execution	20%	65.3%	Developing
Numerical Accuracy	10%	100.0%	Mastery
Presentation & Clarity	10%	49.7%	Emerging

Per-question score formula: $Q = 0.40 \times \text{CU} + 0.20 \times \text{AM} + 0.20 \times \text{SS} + 0.10 \times \text{NA} + 0.10 \times \text{PR}$, where each dimension is scored on a continuous 0–1 scale (full range used, not coarse buckets). Maximum 1.00 mark per question; aggregate over 6 questions.



Concept Dependency Map

Which underlying concepts each question tests, colored by your performance. Green = full, Amber = partial, Red = failed, Grey = not tested.

Concept

	1	2	3	4	5	6
Injectivity via direct contradiction	Amber				Full	
Surjectivity / range computation			Full			
Greatest-integer manipulation		Full				
Case-split on $ x $ or sign			Full			
Combinatorial counting of functions				Full		
Reflection of graphs about a horizontal line						Full

Summary of Rubric Matrix

Each question scored on the same 5 dimensions used for the aggregate. Cell color signals performance: green $\geq 75\%$, amber 50–74%, red $< 50\%$.

Q. NO.	CONCEPTUAL UNDERSTANDING	APPROACH & METHOD	STEP-BY-STEP EXECUTION	NUMERICAL ACCURACY	PRESENTATION & CLARITY	AVERAGE SCORE
6	91%	88%	82%	100%	71%	86%
3	86%	78%	66%	100%	54%	77%
2	82%	76%	68%	100%	47%	75%
4	83%	79%	62%	100%	48%	74%
1	78%	71%	63%	100%	42%	71%
5	74%	58%	51%	100%	36%	64%



Misconceptual Observations Identified

Confusing the periodic graph of $\sin^{-1}(\sin x)$ with its behaviour on a restricted domain. (seen in Q1 option (B))

What's happening: Student sees $\sin^{-1}(\sin x)$ and immediately recalls the triangular-wave graph over all of $\mathbb{R}$, concluding 'not one-one'.

Correct mental model: On the given domain $[-1, 1]$, $|x| < \pi/2$, so $\sin^{-1}(\sin x) = x$ and the function reduces to $\frac{2x}{\pi}$. This IS one-one; the failure is non-surjectivity, since the range $[-2/\pi, 2/\pi]$ is a proper subset of $[-1, 1]$. Always check what the domain restriction does to the formula before invoking the global graph.

Treating an inequality bound as forcing monotonicity rather than just injectivity. (seen in Q5)

What's happening: Because $|f(x) - f(y)| \geq |e^x - e^y|$, f must be monotone, and we must do case analysis on which branch.

Correct mental model: The inequality directly gives injectivity by setting $f(x) = f(y)$ and noting $0 \geq |e^x - e^y| \Rightarrow x = y$. Monotonicity is a downstream consequence of (injectivity + continuity), not the route to it. Use the strongest available hypothesis at the lowest possible level of machinery.

Writing 0/0 where the numerator is identically zero. (seen in Q3, page 7)

What's happening: When both numerator and denominator are expressions in x , the value is 0/0 regardless of whether they are identically zero or just simultaneously zero at a point.

Correct mental model: If the numerator is the zero function on a region (here $e^x - e^{|x|} = 0$ for all $x \geq 0$), then f is identically zero on that region - no indeterminate form. 0/0 is a limit phenomenon at an isolated point, not a description of a function on a set.

Not explicitly verifying which endpoints of a limiting range are attained. (seen in Q3 - $(-1, 0]$ asserted but the 'open at -1 ' justification is implicit.)

What's happening: Compute lim at the boundary, declare that value as the boundary of the range, move on.

Correct mental model: State whether the limit is achieved or merely approached. Here $f \rightarrow -1$ as $x \rightarrow -\infty$ but no finite x gives $f(x) = -1$, so $-1 \notin \text{Range}(f)$, hence the open bracket.



SWOT Analysis

STRENGTHS

- **Correct final answers across the board.** Q1 $\{B, C, D\}$, Q2 $\{A, B\}$, Q3 $(-1, 0]$, Q4 19, Q5 B , Q6 8.5 - all six match the key.
- **Strong reflection-geometry intuition.** Q6 was solved cleanly via $g(x) = 2k - f(x)$ with a supporting diagram.
- **Comfort with case-splits on $|x|$ and on $[x]$.** Q2 and Q3 both required separating regimes; you executed the right partition each time.

OPPORTUNITIES

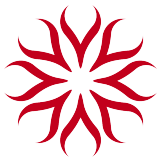
- **Separate injectivity and surjectivity into two explicit checks.** Q1(B) confusion would have been caught instantly if 'one-one?' and 'onto?' were two separate boxes per option.
- **Lead with diagrams.** Q6's diagram made the answer almost trivial - the same habit on Q1(B) and Q5 would have shortened both.

WEAKNESSES

- **Reasoning for option (B) in Q1 is wrong despite the correct letter.** On $[-1, 1]$, $\frac{2}{\pi} \sin^{-1}(\sin x) = \frac{2x}{\pi}$ which is one-one; you labelled it 'not one-one' using a periodic graph that is irrelevant on this restricted domain. The true failure is non-onto.
- **Over-engineered proofs when a one-liner exists.** Q5: injectivity follows in 2 lines from $f(x) = f(y) \Rightarrow 0 \geq |e^x - e^y|$. You spent two pages on monotonicity cases instead.
- **Stray $0/0$ and unjustified limit claims.** Q3 page 7 writes $f(x) = 0/0$ where the numerator is identically zero. Q3 also asserts -1 as the boundary without explicitly showing it is not attained.

THREATS

- **Right-answer-for-wrong-reason patterns can collapse under JEE Advanced scrutiny.** Q1(B) and Q5 both reached the correct conclusion via flawed reasoning; subjective papers would dock heavily even with the right letter.
- **Time cost of over-long arguments.** Q5 alone consumed roughly two notebook pages for a 3-line argument - on a timed paper this is a real risk.



Areas of Improvement

PRIORITY

01

Critical

Audit your reasoning, not just your answer letter.

- ACTION** On the next sitting, after each MCQ, write one sentence answering 'why is this option correct and the others wrong?' Specifically separate the injectivity test from the surjectivity test on every bijection question.
- MEASURE** Zero questions where the correct letter is supported by an incorrect or partially incorrect justification - graded by re-reading your own working before submission.
- BY WHEN** Starting your very next practice attempt.

PRIORITY

02

Important

Use the lightest tool that works.

- ACTION** Before launching into case analysis or auxiliary-angle rewrites, ask: 'Can a single substitution or counter-example settle this?' Q5 (set $f(x) = f(y)$) and Q1C ($x = \pm 1$) are exactly this pattern.
- MEASURE** Average working length per question drops by roughly one notebook page on questions of this difficulty.
- BY WHEN** Within the next two practice assignments.

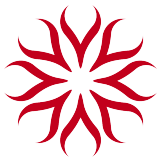
PRIORITY

03

Nice to have

Always state endpoint attainment explicitly.

- ACTION** For every range / domain answer, write a one-line check: 'Is the boundary value actually achieved? If only approached, use open bracket.' Apply this to Q3-style problems systematically.
- MEASURE** Range / domain answers always carry the correct open/closed brackets with a one-line justification beside them.
- BY WHEN** Within the next two weeks of practice.



Per-Question Evaluation

Each question is worth 1 mark. The 5-dimension rubric is applied to every question and weighted to a single mark using the formula on Page 1. Each box below shows the percentage scored on that dimension (big number) with the weighted contribution to the question total below (weight × score). Sub-scores per dimension: 1 (full), 0.5 (partial), 0 (none).

Q1.

0.72 / 1.00

Identify which of $\sin(\sin^{-1} x)$, $\frac{2}{\pi} \sin^{-1}(\sin x)$, $\sin(x) \ln(e^x)$, $x^3 \operatorname{sgn}(x)$ are not bijective on $[-1, 1] \rightarrow [-1, 1]$

Topic: Bijection check on $[-1, 1] \rightarrow [-1, 1]$

Scan Quality: **Acceptable**

YOUR ANSWER

EXPECTED

B, C, D

B, C, D

Conceptual
Understanding

78%

Weight = 0.40 Score = 0.31

Approach
& Method

71%

Weight = 0.20 Score = 0.14

Step-by-Step
Execution

63%

Weight = 0.20 Score = 0.13

Numerical
Accuracy

100%

Weight = 0.10 Score = 0.10

Presentation
& Clarity

42%

Weight = 0.10 Score = 0.04

Feedback: You landed on B, C, D which is the right set, but your justification for option (B) is shaky - on $[-1, 1]$ the function reduces to $\frac{2x}{\pi}$, which is one-one; its failure is being non-onto since the range is $[-2/\pi, 2/\pi] \subsetneq [-1, 1]$. Get into the habit of separating the injectivity test from the surjectivity test and writing one explicit verdict per option. For (C) a single pair $f(1) = f(-1) = \sin 1$ kills it instantly - prefer such counter-examples over symbolic rewrites.

Scan note: Pages 1-3 have a thumb in frame and crowded handwriting; (B)'s reasoning block is partially illegible.

Q2. $f: \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = [x]^2 + [x+1] - 3$

0.76 / 1.00

Topic: Greatest-integer function, many-one / onto

Scan Quality: **Acceptable**

YOUR ANSWER

EXPECTED

A, B

A, B

Conceptual
Understanding

82%

Weight = 0.40 Score = 0.33

Approach
& Method

76%

Weight = 0.20 Score = 0.15

Step-by-Step
Execution

68%

Weight = 0.20 Score = 0.14

Numerical
Accuracy

100%

Weight = 0.10 Score = 0.10

Presentation
& Clarity

47%

Weight = 0.10 Score = 0.05

Feedback: Correct answer and the core idea - f is constant on each $[n, n+1)$ and skips integers like 1 - is exactly right. Trim the working: once you have $([x]+2)([x]-1)$, the zero-set and the missed-integer argument each take two lines. The sign-analysis interval $(-2, 1)$ is a distraction here.

Scan note: Page 5 has a thumb in frame; some intermediate lines are partially cut off.



Q3. $f(x) = \frac{e^x - e^{|x|}}{10e^x + e^{|x|}}, f: \mathbb{R} \rightarrow Y$ onto

0.79 / 1.00Topic: Range of $\frac{e^x - e^{|x|}}{10e^x + e^{|x|}}$ Scan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
$Y = (-1, 0]$	$Y = (-1, 0]$

Conceptual Understanding 86% Weight = 0.40 Score = 0.34	Approach & Method 78% Weight = 0.20 Score = 0.16	Step-by-Step Execution 66% Weight = 0.20 Score = 0.13	Numerical Accuracy 100% Weight = 0.10 Score = 0.10	Presentation & Clarity 54% Weight = 0.10 Score = 0.05
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Feedback: Right answer and right strategy (separate the two cases based on $|x|$). Watch the 0/0 slip on page 7 - when the numerator is identically zero, f is zero, not indeterminate. Also state explicitly that -1 is a limit but never attained, which is what makes the interval open at -1 and closed at 0 .

Scan note: Pages 7-8 are dim; some exponents are hard to read.

Q4. $|C| = |\{f: A \rightarrow B \mid 2 \in f(A), f \text{ not one-one}\}|$ with $|A| = 3, |B| = 4$

0.76 / 1.00Topic: Counting functions: $2 \in f(A)$ and f not one-oneScan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
19	19

Conceptual Understanding 83% Weight = 0.40 Score = 0.33	Approach & Method 79% Weight = 0.20 Score = 0.16	Step-by-Step Execution 62% Weight = 0.20 Score = 0.12	Numerical Accuracy 100% Weight = 0.10 Score = 0.10	Presentation & Clarity 48% Weight = 0.10 Score = 0.05
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Feedback: 19 is right and your instinct to enumerate by image was sound. Next time, pick ONE framework - either 'all functions with 2 in range minus those that are also one-one' ($37 - 18 = 19$) or 'sum over image sizes' - and execute it cleanly. Mixing both invites the kind of overcounting you had to patch with the final subtraction.

Scan note: Page 10 has overwriting on the combinatorial expressions.

**Q5.** $|f(x) - f(y)| \geq |e^x - e^y| \forall x, y$, continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ **0.65** / 1.00Topic: Injectivity from $|f(x) - f(y)| \geq |e^x - e^y|$ Scan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
B (one-one)	B (one-one)

Conceptual Understanding 74% Weight = 0.40 Score = 0.30	Approach & Method 58% Weight = 0.20 Score = 0.12	Step-by-Step Execution 51% Weight = 0.20 Score = 0.10	Numerical Accuracy 100% Weight = 0.10 Score = 0.10	Presentation & Clarity 36% Weight = 0.10 Score = 0.04
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Feedback: Right answer, wrong route. The trivial step is: set $y = x$? no - assume $f(x) = f(y)$, then $0 = |f(x) - f(y)| \geq |e^x - e^y| \geq 0$, forcing $e^x = e^y$, i.e. $x = y$. That's the entire proof of injectivity in three lines. Your monotonicity case-analysis is over-engineered and not even watertight. Also engage with the other options - why is f NOT necessarily surjective? (Because $|f|$ grows at least like e^x , so values below the asymptote are missed.)

Scan note: Pages 11-12 contain many crossed-out lines; case labels are inconsistent.

Q6.**0.88** / 1.00 $f, g : \mathbb{R} \rightarrow \mathbb{R}$ bijective, mirror images about $4y - 17 = 0$; $h = f + g$, find $h(0)$ Topic: Mirror images about $y = 17/4$ Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$h(0) = 8.5$	$h(0) = 8.5$

Conceptual Understanding 91% Weight = 0.40 Score = 0.36	Approach & Method 88% Weight = 0.20 Score = 0.18	Step-by-Step Execution 82% Weight = 0.20 Score = 0.16	Numerical Accuracy 100% Weight = 0.10 Score = 0.10	Presentation & Clarity 71% Weight = 0.10 Score = 0.07
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Feedback: Cleanest question of the paper - you nailed both the geometry ($g(x) = \frac{17}{2} - f(x)$) and the algebra. Notice how the diagram made the argument almost automatic; carry that habit (sketch first, compute second) back to Q5 where the absence of any picture made you over-case-split.



Closing Note

Ankit, six out of six final answers correct on a warm-up is a real result and shows your instincts on functions are sound. The honest read of the paper, though, is that two of those six were right for the wrong reason - and at JEE Advanced level that's the difference between a confident solve and a lucky one.

Your score signals that you can pattern-match to the right answer but you don't yet consistently audit your own justifications. The strongest habit to build now is reading your own argument back and asking 'does this actually prove what I claimed?'

Where to go next.

- **Next assignment:** Open every bijection question by writing two separate headings - 'Injective?' and 'Surjective?' - and resolve each independently before picking an option. This single discipline would have caught the Q1(B) slip.
- **Adjacent topic:** Composition of functions and inverse functions lean on the same injectivity / surjectivity reasoning you used here - practise them in parallel so the bijection toolkit gets reinforced from multiple angles.
- **Speed target:** Aim for roughly 6-8 minutes per question on this difficulty band; Q5 and Q3 currently eat far more than that and trimming them is your fastest gain.

AI-Assisted Evaluation Disclaimer

This report was generated with AI assistance. AI can make mistakes. If any score, feedback, or scan-quality assessment seems off, please flag it for human review. The Standard Subjective Rubric (40% Conceptual Understanding, 20% Approach & Method, 20% Step-by-Step Execution, 10% Numerical Accuracy, 10% Presentation & Clarity) was applied consistently across all questions. Final responsibility for grading decisions rests with the evaluating teacher.

Evaluated by ThinkingSouls evaluation framework · Standard Subjective Rubric (40/20/20/10/10) · All five dimensions weighted to a single mark per question.

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TRACKING ID

TS-QA-NOF1-20260514-AS-001

Scan the QR code or quote this ID when contacting ThinkingSouls about this evaluation - it identifies the exact submission, evaluation date, and version of the rubric used.





Scan Quality Overview

Across the 6 questions: **Excellent: 0** · **Good: 1** · **Acceptable: 5** · **Poor: 0**

Scan quality affects how reliably the work can be evaluated. Each question's scan quality is logged on its individual card. The tips below help you raise scan quality in future submissions.

Scanning Tips for Future Submissions

Clear scans help your evaluator focus on the mathematics instead of decoding the page. A few small habits make a big difference:

1. Keep fingers and thumbs out of frame. Hold the page from a corner that's outside the camera view, or use a flat surface and weights (books on the corners). Thumbs in the frame block content and make pages look unprofessional.	2. One page per image. Don't capture two notebook pages in one shot. Each question's working should be readable without zooming. If a question spans two pages, that's fine - just take two clean shots.
3. Even, indirect lighting. Avoid direct sunlight or a single overhead lamp - both create shadows or glare strips. Diffuse light from a window (no direct sun) or a desk lamp aimed at the wall works best.	4. Page flat, camera straight overhead. A skewed page distorts the math. Use a clipboard or hardback book underneath to flatten the page, and hold the camera parallel to the page (not at an angle).
5. Use a darker pen. Pencil and very light blue ink fade in scans. A regular blue or black ballpoint or gel pen (0.5 mm or thicker) gives the best contrast.	6. Number every question clearly. Write the question number large at the start of each answer (e.g. Q7). If you skip a question and come back to it, mark it - don't leave the evaluator hunting.
7. Box your final answer. A clear final-answer box at the end of each question saves the evaluator from reading the entire derivation just to find your conclusion.	8. Use a scanner app, not raw photos. Apps like Adobe Scan, Microsoft Lens, or CamScanner auto-correct perspective, sharpen text, and compile pages into a single PDF in question order. Always check page order before sending.
9. Check page order before exporting. After scanning, flip through the PDF once. Make sure pages run Q1 → Q2 → ... → last, with no missing pages and no upside-down or sideways shots.	10. Name the file clearly. Format: <i>BatchName_StudentName_AssignmentCode.pdf</i> . Don't use generic names like "Scan001.pdf" - they get lost.

Specific to your NOF1 submission: Several pages (1-3, 5, 7-8, 10-12) have a thumb in frame, dim lighting, or overlapping crossed-out lines that make intermediate steps hard to read. Try to retake page 7 and pages 11-12 in better light and keep fingers clear of the page.